Fast Fourier Transform in zk-STARK Protocols: Interpolation, Encoding and Proximity Proofs.

Paolo De Marinis

15 October 2025

Table of Contents

1. Introduction
2. Arithmetization
3. Optimization Framework
4. The cryptographic proof

Computational Capacity Asymmetry

- **Need**: offload computations to untrusted parties.

Task: compute $f(x)$

Prover **Verifier**

Untrusted

Claim: $y = f(x)$

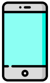
Cryptographic Proof

- **Solution**: requiring a cryptographic proof.

Task: compute $f(x)$

Prover

Verifier



Untrusted

Claim: $y = f(x)$

Verifiable

Cryptographic Proof

Verification Cost, Completeness, Soundness

Verification Cost
Poly-logarithmic compared to re-execution cost T.

Completeness
Honest Prover

$\Downarrow$

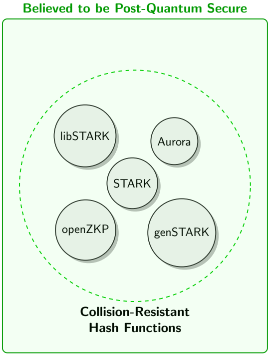
Verifier accepts

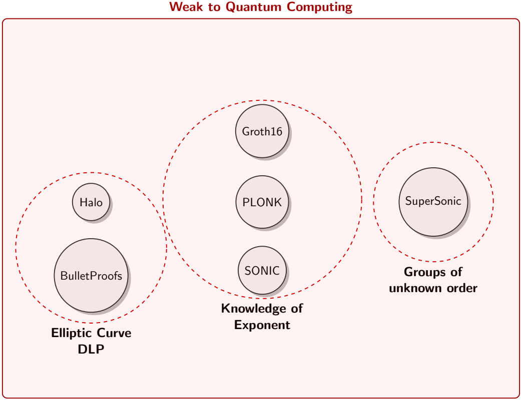
Soundness
Corrupted Prover

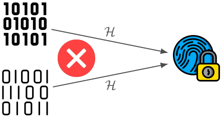
$\Downarrow$

Verifier rejects with **high probability**

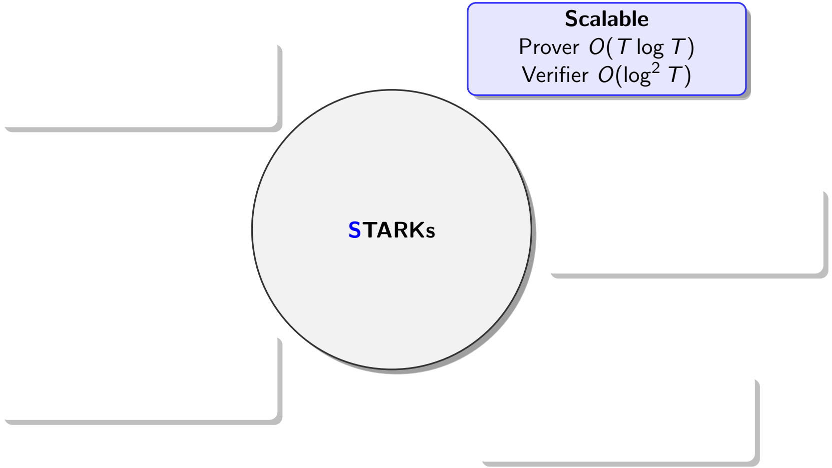
Post-Quantum Security



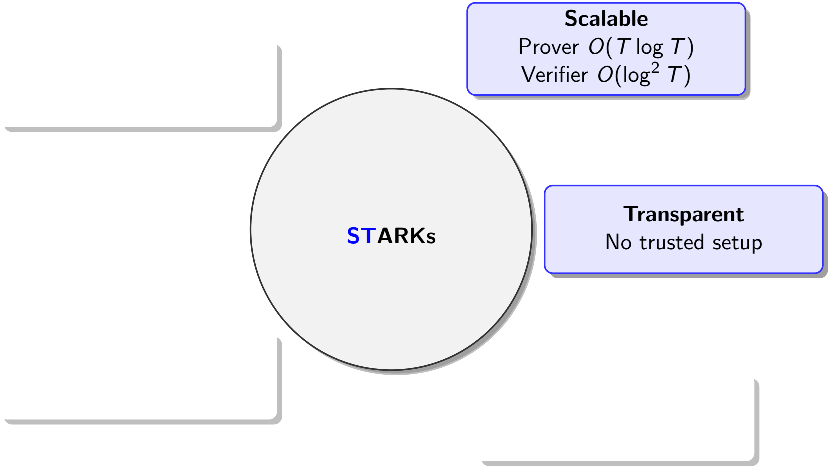




zk-STARK protocols



zk-STARK protocols



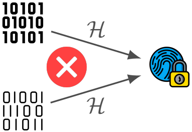
zk-STARK protocols

STARKs

Scalable
Prover $O(T \log T)$
Verifier $O(\log^2 T)$

Transparent
No trusted setup

ARguments
Prover is unable
to find a collision



zk-STARK protocols

STARKs

Scalable
Prover $O(T \log T)$
Verifier $O(\log^2 T)$

Transparent
No trusted setup

ARguments
Prover is unable
to find a collision

of Knowledge
Have the Knowledge
Soundness property

Verifier ✓ $\Longrightarrow$ Prover knows a valid witness

zk-STARK protocols

Zero Knowledge
No leak about witness

Scalable
Prover $O(T \log T)$
Verifier $O(\log^2 T)$

zk-**STARK**s

Transparent
No trusted setup

of Knowledge
Have the Knowledge
Soundness property

ARguments
Prover is unable
to find a collision

The Engine of STARKs

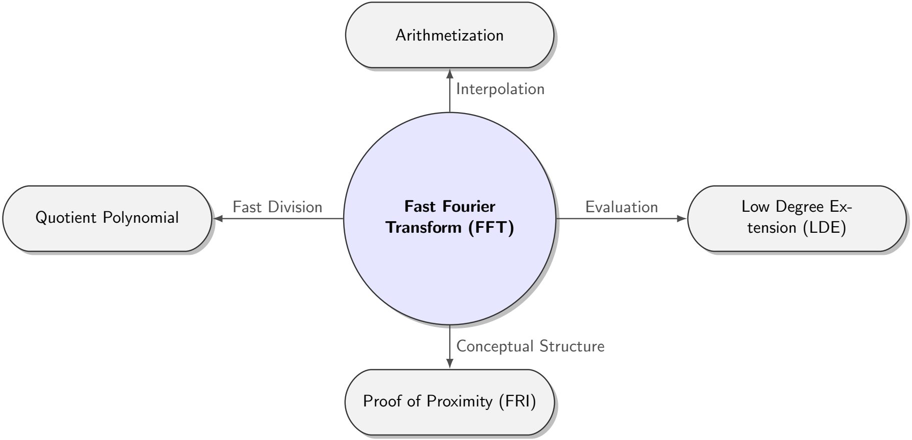


Table of Contents

1 Introduction

2 Arithmetization

3 Optimization Framework

4 The cryptographic proof

Finite State Machine: Fibonacci Sequence in $\mathbb{F}_{17}$

The computation rules

State $s(t)$

$$s(t) = (F_t, F_{t+1}) \in \mathbb{F}_{17}^2$$

Initial state $s(0)$

$$s(0) = (0, 1) \in \mathbb{F}_{17}^2$$

State transition function δ

$$\begin{aligned} \delta \colon \mathbb{F}_{17}^2 &\to \mathbb{F}_{17}^2 \\ (x_1, x_2) &\mapsto (x_2, x_1 + x_2) \end{aligned}$$

Finite State Machine: Fibonacci Sequence in $\mathbb{F}_{17}$

The computation rules

State $s(t)$

$$s(t) = (F_t, F_{t+1}) \in \mathbb{F}_{17}^2$$

Initial state $s(0)$

$$s(0) = (0, 1) \in \mathbb{F}_{17}^2$$

State transition function δ

$$\begin{aligned} \delta \colon \mathbb{F}_{17}^2 &\to \mathbb{F}_{17}^2 \\ (x_1, x_2) &\mapsto (x_2, x_1 + x_2) \end{aligned}$$

Steps T

$$T = 3$$

Final state $s(T)$

$$s(3) = (2, 3)$$

Execution Trace: Fibonacci Sequence in $\mathbb{F}_{17}$

The computation

Computing F_4:

Initial state: $(0, 1) = (F_0, F_1)$

Computing F_4:

Initial state: $(0, 1) = (F_0, F_1)$

Step 1: $\xrightarrow{\delta} (1, 1) = (F_1, F_2)$

Computing F_4:

$$
\begin{aligned}
&\text{Initial state:} && (0,1) = (F_0, F_1) \\
&\text{Step 1:} && \xrightarrow{\delta} (1,1) = (F_1, F_2) \\
&\text{Step 2:} && \xrightarrow{\delta} (1,2) = (F_2, F_3)
\end{aligned}
$$

Computing F_4:

$$\begin{aligned} &\text{Initial state: } (0,1) = (F_0, F_1) \\ &\text{Step 1: } \xrightarrow{\delta} (1,1) = (F_1, F_2) \\ &\text{Step 2: } \xrightarrow{\delta} (1,2) = (F_2, F_3) \\ &\text{Step 3: } \xrightarrow{\delta} (2,3) = (F_3, F_4) \end{aligned}$$

We obtain the **execution trace**

$$x = \begin{pmatrix} x_1(0) & x_2(0) \\ x_1(1) & x_2(1) \\ x_1(2) & x_2(2) \\ x_1(3) & x_2(3) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 3 \end{pmatrix} \in \mathbb{F}_{17}^{4\times 2}$$

From Execution Trace to Trace Polynomials

$$x = \begin{pmatrix} \boxed{\begin{matrix} 0 \\ 1 \\ 1 \\ 2 \end{matrix}} & \boxed{\begin{matrix} 1 \\ 1 \\ 2 \\ 3 \end{matrix}} \end{pmatrix}$$

From Execution Trace to Trace Polynomials

Interpolating over the *evaluation domain* $H = \langle 4 \rangle = \{1, 4, 16, 13\} < \mathbb{F}_{17}^*$

$$x = \left(\boxed{\begin{matrix} 0 \\ 1 \\ 1 \\ 2 \end{matrix}} \boxed{\begin{matrix} 1 \\ 1 \\ 2 \\ 3 \end{matrix}} \right)$$

$$13 \cdot \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 13 & 16 & 4 \\ 1 & 16 & 1 & 16 \\ 1 & 4 & 16 & 13 \end{pmatrix} \left(\boxed{\begin{matrix} 0 \\ 1 \\ 1 \\ 2 \end{matrix}} \right) = \begin{pmatrix} 1 \\ 5 \\ 8 \\ 3 \end{pmatrix} \qquad 13 \cdot \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 13 & 16 & 4 \\ 1 & 16 & 1 & 16 \\ 1 & 4 & 16 & 13 \end{pmatrix} \left(\boxed{\begin{matrix} 1 \\ 1 \\ 2 \\ 3 \end{matrix}} \right) = \begin{pmatrix} 6 \\ 6 \\ 4 \\ 2 \end{pmatrix}$$

From Execution Trace to Trace Polynomials

Interpolating over the *evaluation domain* $H = \langle 4 \rangle = \{1, 4, 16, 13\} < \mathbb{F}_{17}^*$

$$x = \begin{pmatrix} \boxed{\begin{matrix} 0 \\ 1 \\ 1 \\ 2 \end{matrix}} & \boxed{\begin{matrix} 1 \\ 1 \\ 2 \\ 3 \end{matrix}} \end{pmatrix}$$

$$13 \cdot \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 13 & 16 & 4 \\ 1 & 16 & 1 & 16 \\ 1 & 4 & 16 & 13 \end{pmatrix} \begin{pmatrix} \boxed{\begin{matrix} 0 \\ 1 \\ 1 \\ 2 \end{matrix}} \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 8 \\ 3 \end{pmatrix} \qquad 13 \cdot \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 13 & 16 & 4 \\ 1 & 16 & 1 & 16 \\ 1 & 4 & 16 & 13 \end{pmatrix} \begin{pmatrix} \boxed{\begin{matrix} 1 \\ 1 \\ 2 \\ 3 \end{matrix}} \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \\ 4 \\ 2 \end{pmatrix}$$

$$\boxed{W_1(X) = 3X^3 + 8X^2 + 5X + 1} \qquad \boxed{W_2(X) = 2X^3 + 4X^2 + 6X + 6}$$

From Computational Integrity Statement to Algebraic Intermediate Representation (AIR)

Interpolating over the *evaluation domain*

$$H = \{h_0, h_1, h_2, h_3\} = \{1, 4, 16, 13\}$$

$$x = \left(\begin{pmatrix} 0 \\ 1 \\ 1 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 2 \\ 3 \end{pmatrix} \right) = \left(\begin{pmatrix} W_1(h_0) \\ W_1(h_1) \\ W_1(h_2) \\ W_1(h_3) \end{pmatrix} \begin{pmatrix} W_2(h_0) \\ W_2(h_1) \\ W_2(h_2) \\ W_2(h_3) \end{pmatrix} \right)$$

Border condition on $s(T)$

$$s_2(3) = 3$$

Border conditions on $s(0)$

$$s_1(0) = 0, s_2(0) = 1$$

Transition Conditions

$$s_1(t+1) = s_2(t)$$
$$s_2(t+1) = s_1(t) + s_2(t)$$

From Computational Integrity Statement to Algebraic Intermediate Representation (AIR)

Interpolating over the *evaluation domain*

$$H = \{h_0, h_1, h_2, h_3\} = \{1, 4, 16, 13\}$$

$$x = \left(\begin{array}{|c|c|} 0 & 1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 3 \end{array}\right) = \left(\begin{array}{|c|c|} W_1(h_0) & W_2(h_0) \\ W_1(h_1) & W_2(h_1) \\ W_1(h_2) & W_2(h_2) \\ W_1(h_3) & W_2(h_3) \end{array}\right)$$

Border constraint on $h_3 = 13$

$$W_2(h_3) - 3 = 0$$

Border conditions on $s(0)$

$$s_1(0) = 0, s_2(0) = 1$$

Transition Conditions

$$s_1(t+1) = s_2(t)$$

$$s_2(t+1) = s_1(t) + s_2(t)$$

From Computational Integrity Statement to Algebraic Intermediate Representation (AIR)

Interpolating over the *evaluation domain*

$$H = \{h_0, h_1, h_2, h_3\} = \{1, 4, 16, 13\}$$

$$x = \left(\begin{array}{c|c} 0 & 1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 3 \end{array}\right) = \left(\begin{array}{c|c} W_1(h_0) & W_2(h_0) \\ W_1(h_1) & W_2(h_1) \\ W_1(h_2) & W_2(h_2) \\ W_1(h_3) & W_2(h_3) \end{array}\right)$$

Border constraint on $h_3 = 13$

$$W_2(h_3) - 3 = 0$$

Border constraint on $h_0 = 1$

$$W_1(h_0) = 0, W_2(h_0) - 1 = 0$$

Transition Conditions

$$s_1(t+1) = s_2(t)$$

$$s_2(t+1) = s_1(t) + s_2(t)$$

From Computational Integrity Statement to Algebraic Intermediate Representation (AIR)

Interpolating over the *evaluation domain*

$$H = \{h_0, h_1, h_2, h_3\} = \{1, 4, 16, 13\}$$

$$x = \left(\boxed{\begin{matrix}0\\1\\1\\2\end{matrix}}\boxed{\begin{matrix}1\\1\\2\\3\end{matrix}}\right) = \left(\boxed{\begin{matrix}W_1(h_0)\\W_1(h_1)\\W_1(h_2)\\W_1(h_3)\end{matrix}}\boxed{\begin{matrix}W_2(h_0)\\W_2(h_1)\\W_2(h_2)\\W_2(h_3)\end{matrix}}\right)$$

Border constraint on $h_3 = 13$

$$W_2(h_3) - 3 = 0$$

Border constraint on $h_0 = 1$

$$W_1(h_0) = 0, W_2(h_0) - 1 = 0$$

Transition constraints for $h_t \in H \setminus \{h_3\}$

$$W_1(h_{t+1}) - W_2(h_t) = 0$$

$$W_2(h_{t+1}) - (W_1(h_t) + W_2(h_t)) = 0$$

From Computational Integrity Statement to Algebraic Intermediate Representation (AIR)

Interpolating over the *evaluation domain*

$$H = \{h_0, h_1, h_2, h_3\} = \{1, 4, 16, 13\}$$

$$x = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} W_1(h_0) & W_2(h_0) \\ W_1(h_1) & W_2(h_1) \\ W_1(h_2) & W_2(h_2) \\ W_1(h_3) & W_2(h_3) \end{pmatrix}$$

Border constraint on $h_3 = 13$

$$W_2(h_3) - 3 = 0$$

Border constraint on $h_0 = 1$

$$W_1(h_0) = 0, W_2(h_0) - 1 = 0$$

Transition constraints for $h_t \in H \setminus \{h_3\}$ and $\sigma(X) = 4X$

$$W_1(\sigma(h_t)) - W_2(h_t) = 0$$

$$W_2(\sigma(h_t)) - (W_1(h_t) + W_2(h_t)) = 0$$

From Constraints to Constraint Polynomials

Transition constraints for $h_t \in H \setminus \{h_3\}$ and $\sigma(X) = 4X$

$$W_1(\sigma(h_t)) - W_2(h_t) = 0$$
$$W_2(\sigma(h_t)) - (W_1(h_t) + W_2(h_t)) = 0$$

From Constraints to Constraint Polynomials

Transition constraints for $h_t \in H \setminus \{h_3\}$ and $\sigma(X) = 4X$

$$\mathrm{Check}_1(h_t) := W_1(\sigma(h_t)) - W_2(h_t) = 0$$
$$\mathrm{Check}_2(h_t) := W_2(\sigma(h_t)) - (W_1(h_t) + W_2(h_t)) = 0$$

From Constraints to Constraint Polynomials

Transition constraints for $h_t \in H \setminus \{h_3\}$ and $\sigma(X) = 4X$

$\text{Check}_1(h_t) := P_1(\mathbf{W}(h_t), \mathbf{W}(\sigma(h_t))) = 0 \quad$ where $P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - X_2$

$\text{Check}_2(h_t) := W_2(\sigma(h_t)) - (W_1(h_t) + W_2(h_t)) = 0$

From Constraints to Constraint Polynomials

Transition constraints for $h_t \in H \setminus \{h_3\}$ **and** $\sigma(X) = 4X$

$$\text{Check}_1(h_t) := P_1(\mathbf{W}(h_t), \mathbf{W}(\sigma(h_t))) = 0 \quad \text{where } P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - X_2$$

$$\text{Check}_2(h_t) := P_2(\mathbf{W}(h_t), \mathbf{W}(\sigma(h_t))) = 0 \quad \text{where } P_2(\mathbf{X}, \mathbf{Y}) = Y_2 - (X_1 + X_2)$$

$$\Downarrow$$

Algbraic Intermediate Representation

$$\mathcal{AIR}(\mathcal{A}) = (\mathbb{F}, T, w, H, \sigma(X), \mathcal{P}_\delta, \mathcal{B})$$

Solutions Sets

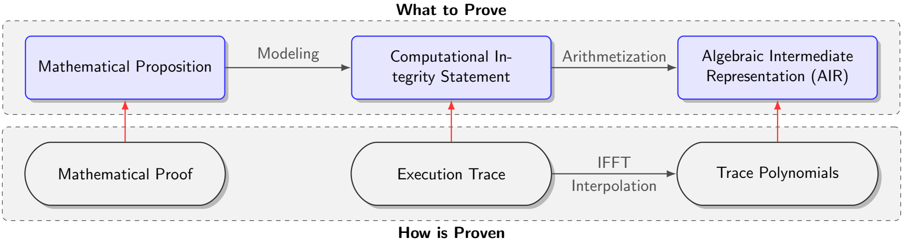
$$\mathcal{W}_{\mathcal{AIR}(\mathcal{A})} := \{\mathbf{W}(X) \in \mathbb{F}[X]^w \mid \mathbf{W}(X) \text{ satisfies } \mathcal{AIR}(\mathcal{A})\} \neq \emptyset$$

$$\Updownarrow$$

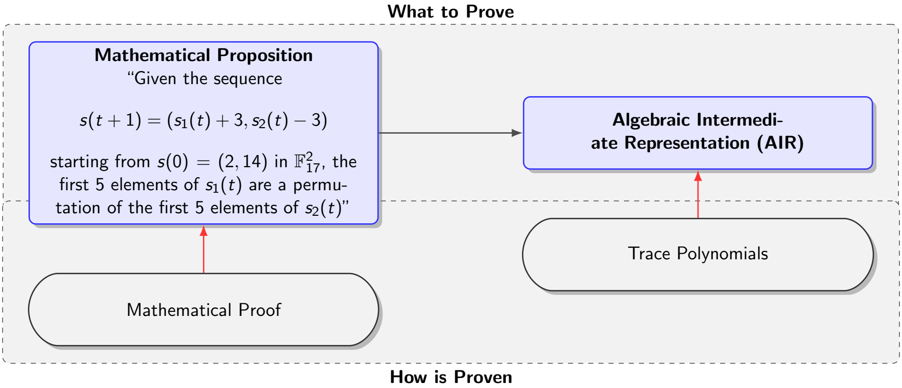
$$\mathcal{T}_{\mathcal{A}} := \{x \in \mathbb{F}_q^{(T+1)\times w} \mid x(t) = \mathbf{W}(h_t) \text{ for } h_t \in H \text{ and } \mathbf{W} \in \mathcal{W}_{\mathcal{AIR}(\mathcal{A})}\} \neq \emptyset$$

- **Goal:** convince the Verifier that $\mathcal{W}_{\mathcal{AIR}(\mathcal{A})} \neq \emptyset$

Overview


What to Prove
Mathematical Proposition
Modeling
Computational Integrity Statement
Arithmetization
Algebraic Intermediate Representation (AIR)
Mathematical Proof
Execution Trace
IFFT
Interpolation
Trace Polynomials
How is Proven

Proving Permutations



AIR and permutations

Execution Trace

$$x = \begin{pmatrix} 2 & 14 \\ 5 & 11 \\ 8 & 8 \\ 11 & 5 \\ 14 & 2 \end{pmatrix} \in \mathbb{F}_{17}^{5\times 2}$$

Initial State

$$s(0) = (2, 14)$$

State transition function δ

$$\begin{aligned} \delta \colon \mathbb{F}_{17}^2 &\to \mathbb{F}_{17}^2 \\ (x_1, x_2) &\mapsto (x_1 + 3, x_2 - 3) \end{aligned}$$

AIR and permutations

Execution Trace

$$x = \begin{pmatrix} 2 & 14 \\ 5 & 11 \\ 8 & 8 \\ 11 & 5 \\ 14 & 2 \end{pmatrix} \in \mathbb{F}_{17}^{5\times 2}$$

State Constraint Polynomials

$$(B_1(\mathbf{X}) = X_1 - 2, \{h_0\})$$
$$(B_2(\mathbf{X}) = X_2 - 14, \{h_0\})$$

Local Constraint Polynomials

$$(P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - (X_1 + 3), H \setminus \{h_4\})$$
$$(P_2(\mathbf{X}, \mathbf{Y}) = Y_2 - (X_2 - 3), H \setminus \{h_4\})$$

AIR and permutations

Execution Trace

$$x = \begin{pmatrix} 2 & 14 \\ 5 & 11 \\ 8 & 8 \\ 11 & 5 \\ 14 & 2 \end{pmatrix} \in \mathbb{F}_{17}^{5\times 2}$$

Local Constraint Polynomials

$$(P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - (X_1 + 3), H \setminus \{h_4\})$$
$$(P_2(\mathbf{X}, \mathbf{Y}) = Y_2 - (X_2 - 3), H \setminus \{h_4\})$$

State Constraint Polynomials

$$(B_1(\mathbf{X}) = X_1 - 2, \{h_0\})$$
$$(B_2(\mathbf{X}) = X_2 - 14, \{h_0\})$$

Non-local Constraint Polynomial

$$(P_{\pi_1}(\mathbf{X}, \mathbf{Y}) = Y_2 - X_1, H)$$

where $\pi_1 : H \to H$ is a bijection

An *AIR* is a tuple

$$\mathcal{AIR} = (\mathbb{F}, T, w, H, \sigma(X), \mathcal{P}, \mathcal{P}_{\boldsymbol{\pi}}, \mathcal{B})$$

where

- $\mathcal{P} = \{(P_1, H_{P_1}), \ldots, (P_s, H_{P_s})\}$
- $\mathcal{P}_{\boldsymbol{\pi}} = \{(P_{\pi_1}, H_{P_{\pi_1}}), \ldots, (P_{\pi_r}, H_{P_{\pi_r}})\}$ and $\pi_k : H \to H$ is a bijection.
- $\mathcal{B} = \{(B_1, H_{B_1}\}), \ldots, (B_m, H_{B_m})\}$

Polynomial Witness

$(\mathbf{W}(X), \boldsymbol{\pi}(X))$

Witness Sets

$$\mathcal{W}_{\mathcal{AIR}} := \{(\mathbf{W}(X), \boldsymbol{\pi}(X)) \in \mathbb{F}[X]^w \times \mathbb{F}[X]^r \mid (\mathbf{W}(X), \boldsymbol{\pi}(X)) \text{ satisfy } \mathcal{AIR}\}.$$

$$\mathcal{T}_{\mathcal{AIR}} := \{(x, p) \in \mathbb{F}^{(T+1)\times w} \times H^{(T+1)\times r} \mid \\ (x(t), p(t)) = (\mathbf{W}(h_t), \boldsymbol{\pi}(h_t)) \;\text{ for } h_t \in H \text{ and } (\mathbf{W}, \boldsymbol{\pi}) \in \mathcal{W}_{\mathcal{AIR}}\}$$

Table of Contents

1 Introduction

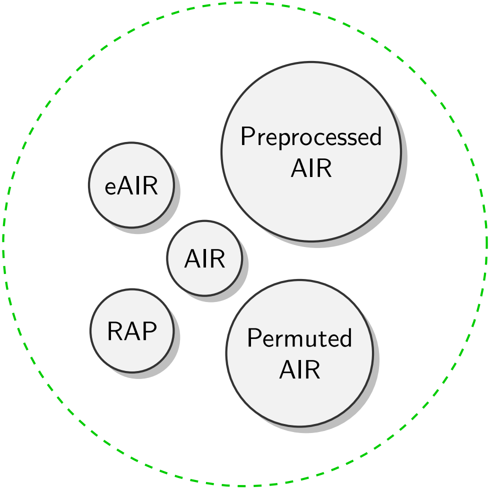
2 Arithmetization

3 Optimization Framework

4 The cryptographic proof

Problems in Literature

Lack of an Optimization Framework



What do they have in common?

Preorder Relations

Valid Extension

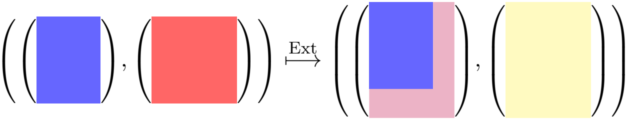
$\mathcal{AIR}_1 \preceq \mathcal{AIR}_2$

Probabilistically Sound Extension

$\mathcal{AIR}_1 \preceq^{\epsilon}_{\mathsf{Pr}} \mathcal{AIR}_2$

Valid Extension

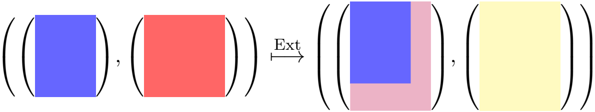
$$\mathcal{AIR}_1 \preceq \mathcal{AIR}_2$$

- $(x, p) =$  $= (x', p')$
- $(x, p) \in \mathcal{T}_{\mathcal{AIR}_1} \iff \mathrm{Ext}(x, p) \in \mathcal{T}_{\mathcal{AIR}_2}$

Preorder Relations: Probabilistically Sound Extension

Probabilistically Sound Extension

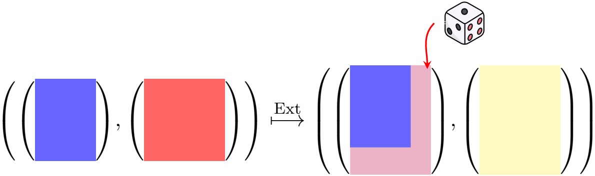
$$\mathcal{AIR}_1 \preceq^{\epsilon}_{\mathsf{Pr}} \mathcal{AIR}_2$$

- $((x, p), R) =$  $= (x', p')$
- $(x, p) \in \mathcal{T}_{\mathcal{AIR}_1} \implies \mathrm{Ext}((x, p), R) \in \mathcal{T}_{\mathcal{AIR}_2}$ for each $R \in \mathbb{F}^s$
- $(x, p) \notin \mathcal{T}_{\mathcal{AIR}_1} \implies \Pr_{R \in \mathbb{F}^s}[\mathrm{Ext}((x, p), R) \in \mathcal{T}_{\mathcal{AIR}_2}] \leq \epsilon$

Preorder Relations: Probabilistically Sound Extension

Probabilistically Sound Extension

$$\mathcal{AIR}_1 \preceq^{\epsilon}_{\mathsf{Pr}} \mathcal{AIR}_2$$

- $((\textcolor{blue}{x}, \textcolor{red}{p}), R) =$  $= (x', p')$
- $(x, p) \in \mathcal{T}_{\mathcal{AIR}_1} \implies \mathrm{Ext}((x, p), R) \in \mathcal{T}_{\mathcal{AIR}_2}$ for each $R \in \mathbb{F}^s$
- $(x, p) \notin \mathcal{T}_{\mathcal{AIR}_1} \implies \Pr_{R \in \mathbb{F}^s}[\mathrm{Ext}((x, p), R) \in \mathcal{T}_{\mathcal{AIR}_2}] \leq \epsilon$

Optimization: Padding

The Extension

$$H = \{h_0, h_1, h_2, h_3, h_4\} \subset \mathbb{F}_{17}$$

$$\Big\downarrow f$$

$$H' = \langle 2 \rangle = \{\underbrace{1, 2, 4, 8, 16}_{\text{Target for H}}, 15, 13, 9\} < \mathbb{F}_{17}^*$$

$$(x, p) = \left(\begin{pmatrix} 2 & 14 \\ 5 & 11 \\ 8 & 8 \\ 11 & 5 \\ 14 & 2 \end{pmatrix}, \begin{pmatrix} h_4 \\ h_3 \\ h_2 \\ h_1 \\ h_0 \end{pmatrix} \right) \overset{\text{Ext}}{\rightarrow} \left(\begin{pmatrix} 2 & 14 \\ 5 & 11 \\ 8 & 8 \\ 11 & 5 \\ 14 & 2 \\ \hdashline * & * \\ * & * \\ * & * \end{pmatrix}, \begin{pmatrix} 16 \\ 8 \\ 4 \\ 2 \\ 1 \\ \hdashline 15 \\ 13 \\ 9 \end{pmatrix} \right) = (x', p')$$

Optimization: Padding

New Polynomial Constraints

$$H = \{h_0, h_1, h_2, h_3, h_4\} \subset \mathbb{F}_{17}$$

$$\Big\downarrow f$$

$$H' = \langle 2 \rangle = \{\underbrace{1, 2, 4, 8, 16}_{\text{Target for H}}, 15, 13, 9\} < \mathbb{F}_{17}^*$$

State Constraint Polynomials

$$(B_1(\mathbf{X}) = X_1 - 2, \{h_0\})$$
$$(B_2(\mathbf{X}) = X_2 - 14, \{h_0\})$$

Local Constraint Polynomials

$$(P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - (X_1 + 3), H \setminus \{h_4\})$$
$$(P_2(\mathbf{X}, \mathbf{Y}) = Y_2 - (X_2 - 3), H \setminus \{h_4\})$$

Non-local Constraint Polynomial

$$(P_{\pi_1}(\mathbf{X}, \mathbf{Y}) = Y_2 - X_1, H)$$

where $\pi_1 : H \to H$ is a bijection

$$H = \{h_0, h_1, h_2, h_3, h_4\} \subset \mathbb{F}_{17}$$

$$\Big\downarrow f$$

$$H' = \langle 2 \rangle = \{\underbrace{1, 2, 4, 8, 16}_{\text{Target for H}}, 15, 13, 9\} < \mathbb{F}_{17}^*$$

State Constraint Polynomials

$$(B_1(\mathbf{X}) = X_1 - 2, \{1\})$$
$$(B_2(\mathbf{X}) = X_2 - 14, \{1\})$$

Local Constraint Polynomials

$$(P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - (X_1 + 3), H \setminus \{h_4\})$$
$$(P_2(\mathbf{X}, \mathbf{Y}) = Y_2 - (X_2 - 3), H \setminus \{h_4\})$$

Non-local Constraint Polynomial

$$(P_{\pi_1}(\mathbf{X}, \mathbf{Y}) = Y_2 - X_1, H)$$

where $\pi_1 : H \to H$ is a bijection

$$H = \{h_0, h_1, h_2, h_3, h_4\} \subset \mathbb{F}_{17}$$

$$\Big\downarrow f$$

$$H' = \langle 2 \rangle = \{\underbrace{1, 2, 4, 8, 16}_{\text{Target for H}}, 15, 13, 9\} < \mathbb{F}_{17}^*$$

State Constraint Polynomials

$$(B_1(\mathbf{X}) = X_1 - 2, \{1\})$$
$$(B_2(\mathbf{X}) = X_2 - 14, \{1\})$$

Local Constraint Polynomials

$$(P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - (X_1 + 3), \{1, 2, 4, 8\})$$
$$(P_2(\mathbf{X}, \mathbf{Y}) = Y_2 - (X_2 - 3), \{1, 2, 4, 8\})$$

Non-local Constraint Polynomial

$$(P_{\pi_1}(\mathbf{X}, \mathbf{Y}) = Y_2 - X_1, H)$$

where $\pi_1 : H \to H$ is a bijection

Optimization: Padding

New Polynomial Constraints

$$H = \{h_0, h_1, h_2, h_3, h_4\} \subset \mathbb{F}_{17}$$

$$\Big\downarrow f$$

$$H' = \langle 2 \rangle = \{\underbrace{1, 2, 4, 8, 16}_{\text{Target for H}}, 15, 13, 9\} < \mathbb{F}_{17}^*$$

State Constraint Polynomials

$$(B_1(\mathbf{X}) = X_1 - 2, \{1\})$$
$$(B_2(\mathbf{X}) = X_2 - 14, \{1\})$$

Local Constraint Polynomials

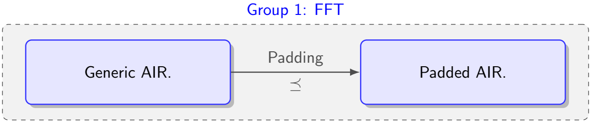
$$(P_1(\mathbf{X}, \mathbf{Y}) = Y_1 - (X_1 + 3), \{1, 2, 4, 8\})$$
$$(P_2(\mathbf{X}, \mathbf{Y}) = Y_2 - (X_2 - 3), \{1, 2, 4, 8\})$$

Non-local Constraint Polynomial

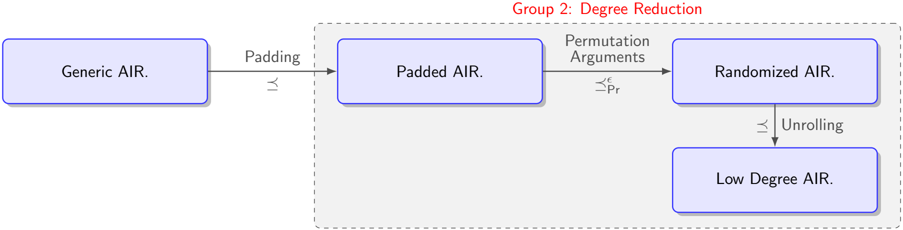
$$(P_{\pi_1}(\mathbf{X}, \mathbf{Y}) = Y_2 - X_1, \{1, 2, 4, 8, 16\})$$

where $\pi_1 : H' \to H'$ is a bijection

Optimizations



Optimizations



Optimizations

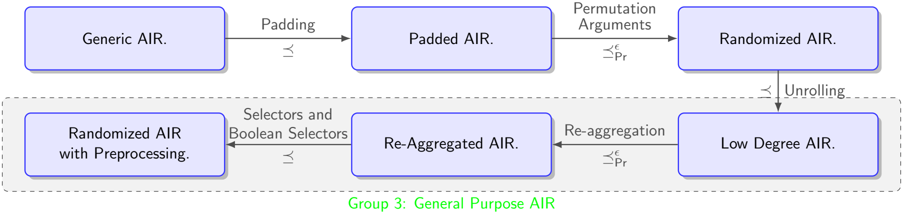


Table of Contents

1. Introduction

2. Arithmetization

3. Optimization Framework

4. The cryptographic proof

Algebraic Linking

Original set of check polynomials

$\text{Check}_i(X)$

$\xrightarrow[\text{Linear Combination}]{\text{Random}}$

Quotient polynomial

$$Q(X) = \sum_i \alpha^{i-1} \frac{\text{Check}_i(X)}{Z_H(X)}$$

$\mathbb{F} = \mathbb{F}_{97}, T + 1 = 8$ and $H = \langle \omega_8 \rangle = \langle 64 \rangle$

$$Q(X) = X^{14} + 2X^{13} + 3X^5 + 14X + 5 \in \mathbb{F}_{97}[X]$$

Algebraic Linking

Original set of check polynomials

$\mathrm{Check}_i(X)$

$\xrightarrow[\text{Linear Combination}]{\text{Random}}$

Quotient polynomial

$$Q(X) = \sum_i \alpha^{i-1} \frac{\mathrm{Check}_i(X)}{Z_H(X)}$$

$\xrightarrow{\text{Splitting}}$

Quotient trace polynomials

$$Q(X) = \sum_{j=1}^{S} X^{j-1} Q_j(X^S)$$

$\mathbb{F} = \mathbb{F}_{97}, T + 1 = 8$ and $H = \langle \omega_8 \rangle = \langle 64 \rangle$

$$Q(X) = X^{14} + 2X^{13} + 3X^5 + 14X + 5 \in \mathbb{F}_{97}[X]$$

With $S = 2$

$$Q(X) = \underbrace{(X^{14} + 5)}_{even} + X \cdot \underbrace{(2X^{12} + 3X^4 + 14)}_{odd} = Q_1(X^2) + X \cdot Q_2(X^2)$$

with

$$Q_1(Y) = Y^7 + 5$$
$$Q_2(Y) = 2Y^6 + 3Y^2 + 14$$

Consistency Check and Low Degree Polynomials

$H = \langle \omega_8 \rangle$, $D = g \cdot \langle \omega_{\beta \cdot 8} \rangle$ and random $z \in \mathbb{F}_{97} \setminus (D \cup H)$

Verifier checks:

$$\sum_i \alpha^{i-1} \frac{\mathsf{Check}_i(z)}{Z_H(z)} = \sum_{j=1}^{2} X^{j-1} Q_j(z^2)$$

Consistency Check and Low Degree Polynomials

$H = \langle \omega_8 \rangle$, $D = g \cdot \langle \omega_{\beta \cdot 8} \rangle$ and random $z \in \mathbb{F}_{97} \setminus (D \cup H)$

Verifier checks:

$$\sum_i \alpha^{i-1} \frac{\mathrm{Check}_i(z)}{Z_H(z)} = \sum_{j=1}^{2} X^{j-1} Q_j(z^2)$$

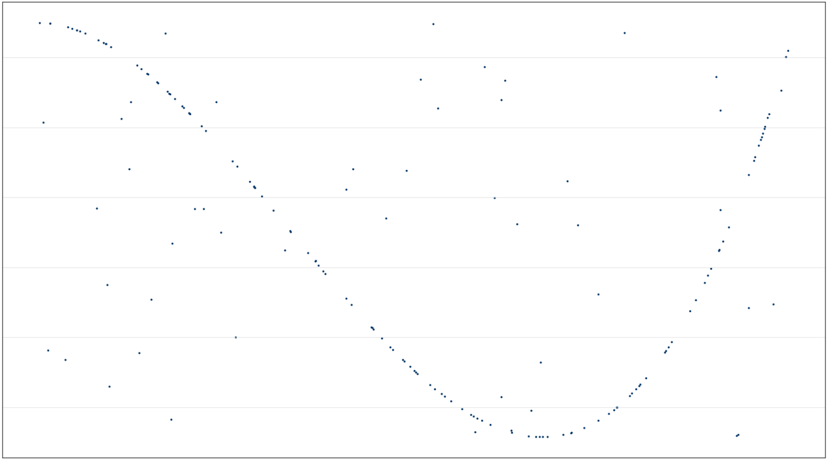
Last check: are those of low degree?

$$F_{j_1,1}(X) = \frac{W_{j_1}(X) - W_{j_1}(z)}{X - z}$$

$$F_{j_1,2}(X) = \frac{W_{j_1}(\sigma(X)) - W_{j_1}(\sigma(z))}{X - z}$$

$$F_j(X) = \frac{Q_j(X) - Q_j(z^2)}{X - z^2}$$

Proximity



FRI Protocol: Split and Fold

PROVER

Split: $F_0(X) = F_{0,even}(X^2) + X \cdot F_{0,odd}(X^2)$

Receive from Verifier: Random value $\lambda_0 \in \mathbb{F}_q$

Fold: $F_1(Y) = F_{0,even}(Y) + \lambda_0 \cdot F_{0,odd}(Y)$

$$\vdots$$

Degree is halved until constant polynomial F_r!

FRI Protocol: Query and Verify

VERIFIER

Queries: $F_0(\eta_0), F_0(-\eta_0), F_1(\eta_0^2)$ for $\eta_0 \in D$

Checks: Consistency with folding relation

$\vdots$

Until constant polynomial F_r!